

The Subspace Portfolio Estimated on the Grassmann Manifold

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Abstract

This document describes a subspace portfolio for monthly construction, estimated on the Grassmann manifold. The model addresses an identification problem in a factor construction by estimating the subspace of the characteristic space that carries the return relevant structure directly, as a point on the Grassmann manifold, and then applying a factor network only within that subspace. We present the identification argument, the manifold parameterisation, the objective that selects the subspace, the Riemannian optimisation, and the separation of the subspace from the nonlinearity.

1 Overview and motivation

The model is built to resolve an identification problem. Consider a factor construction in which a network produces a loading matrix whose columns become long short portfolios, which are then combined into a book. That matrix is not identified. Replacing it by any invertible transform, with the inverse applied to the combination, leaves the traded book essentially unchanged, so the object that the objective can actually distinguish is the subspace that the columns span, not the columns themselves. A network that parameterises the matrix therefore spends capacity choosing a basis to which the objective is nearly blind. The model removes that redundancy by estimating the subspace directly and quotienting out the choice of basis within it.

2 The Grassmann parameterisation

A subspace of dimension k in the characteristic space of dimension p is a point on the Grassmann manifold $Gr(k, p)$. It is represented by an orthonormal frame $Y \in \mathbb{R}^{p \times k}$ with $Y^\top Y = I_k$, understanding that two frames related by a rotation Y and YQ with Q orthogonal represent the same subspace. The redundancy that the loading matrix suffered is thereby made explicit, and it is removed by the geometry of the optimisation rather than learned around.

3 The subspace objective

The subspace is chosen by combining two complementary criteria, both invariant to the level of returns in a month because the response is sliced on its within month rank. The first is the variance of the predictor means across slices of the response, which detects directions on which the response depends monotonically. It is blind to symmetric dependence, since a predictor that enters through an even function has the same mean in high and low slices. To capture such directions the second criterion adds the deviation of each slice covariance from the pooled covariance, which is large exactly where the dependence is symmetric. Writing B for the combined between slice matrix and

W for the within predictor scatter, the subspace maximises the generalised Rayleigh quotient inside it,

$$\max_{Y^\top Y = I} \text{tr} \left[(Y^\top W Y)^{-1} (Y^\top B Y) \right], \quad (1)$$

which is the objective of sliced inverse regression restricted to the subspace.

4 Riemannian ascent

The optimisation respects the manifold structure. At each step the Euclidean gradient G of the objective is projected onto the horizontal space, which removes the directions that lie in the current span and only rotate the frame,

$$G^h = G - Y(Y^\top G). \quad (2)$$

What remains is the Grassmann gradient, the component that genuinely moves the subspace. The iterate is advanced along the normalised horizontal gradient and returned to the manifold by polar retraction, which replaces the updated frame by its nearest orthonormal frame through a singular value decomposition,

$$Y \leftarrow \text{polar} (Y + \alpha G^h / \|G^h\|). \quad (3)$$

Because the redundant directions are projected out at every step, the optimisation never expends effort on the choice of basis and moves only the subspace itself.

5 The subspace and the factor network

The estimated subspace is used so that a separation of roles is achieved, which is the point of the design. Each month the securities are projected onto the subspace, the projections are ranked within the month, and the ranked projections are read by a factor machinery consisting of a loading network conditioned on the cross section through a pooled context and a precision network that supplies an inverse variance weighting, whose factor portfolios are combined by a mean variance rule on their own realised returns. The manifold decides which directions matter and the network decides what to do with them, whereas a single unconstrained loading matrix conflates the two. The combined book is neutralised to market beta without an intercept and scaled to a ten percent annual volatility target against a shrinkage covariance from the trailing daily window, and the subspace and the networks are refitted yearly on months up to that year's cutoff.

6 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 2.842 in the volatility scaled form and 3.046 in the unscaled form. The evidence on the subspace model is most informative through an ablation. A fitted subspace of dimension forty eight achieved a Sharpe ratio of 2.97 with the network and only 1.87 without it, so that removing the network, and with it the pooled cross sectional context, reduced the Sharpe ratio by more than a full point. We can therefore conclude that allowing a security's score to depend on the composition of its contemporaneous cross section is more important than any other single architectural choice tested, and that the manifold, while a principled way to remove the identification redundancy, delivers its value only when paired with the cross sectionally coupled nonlinearity. Within the limitations of the present analysis, the model demonstrates that the useful degrees of freedom are the subspace and the coupling, not the particular basis in which the factors are expressed.